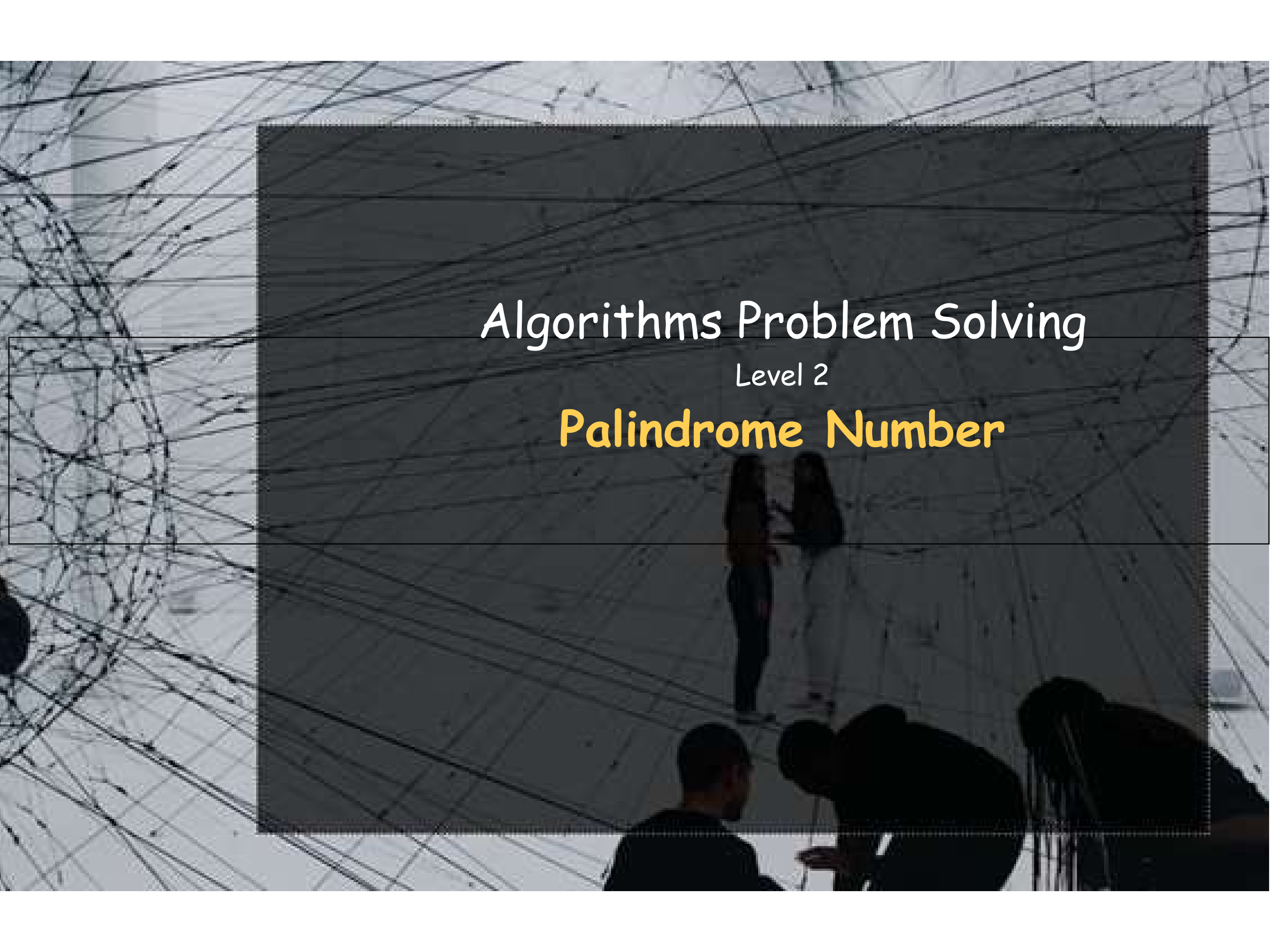


Algorithms Problem Solving

Level 2

Palindrome Number



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Palindrome Number

Definition

- ◆ A Palindrome Number is a number that reads the same from left to right and from right to left.
- ◆ For example, `12321` is a Palindrome because reversing it gives `12321`.

Explanation

- ◆ ``ReadNumber()`` reads a non-negative number from the user.
- ◆ ``ReverseNumber()`` creates the reversed version of the number.
- ◆ ``number % 10`` extracts the last digit.
- ◆ ``number = number / 10`` removes the last digit.
- ◆ ``numb = numb * 10 + remainder`` builds the reversed number digit by digit.
- ◆ ``ReverseNumber()`` returns the reversed number.
- ◆ ``IsPalindrome()`` compares the original number with its reversed version.
- ◆ If both numbers are equal, the number is a Palindrome.
- ◆ ``PrintResult()`` prints the final result.

Steps in main()

1. Read a non-negative number.
2. Reverse the number using `ReverseNumber()`.
3. Compare the original number with the reversed number.
4. If they are equal, return `true`.
5. Otherwise, return `false`.
6. Print whether the number is Palindrome or not.

💡 Example

◆ Input:

`12321`

◆ Reverse the number:

`12321 → 1 → 12 → 123 → 1232 → 12321`

◆ Original number:

`12321`

◆ Reversed number:

`12321`

◆ Since both numbers are equal, it is a Palindrome.

➡ Result

`12321 Yes it is a Palindrome number.`

Note

- ◆ `ReverseNumber()` is reused from the previous lessons.
- ◆ `IsPalindrome()` makes the program easier to understand because it only focuses on the comparison.
- ◆ The expression `numb = numb \* 10 + remainder` is what builds the reversed number.

⊘ Warning

- ◆ `ReadNumber()` allows `0` because it only checks `numb < 0`. Zero is technically a Palindrome Number, so this is valid.
- ◆ The program uses `int`, so very large numbers may exceed the range of the type.



Comparison

Palindrome Number

- ✓ Original number equals its reversed number.
- ✓ Example: `12321 → 12321`.
- ✓ Example: `1221 → 1221`.

Not Palindrome Number

- ✓ Original number is different from its reversed number.
- ✓ Example: `1234 → 4321`.

✦ Summary

- ✓ A Palindrome Number reads the same in both directions.
- ✓ `` % 10`` extracts the last digit.
- ✓ `` / 10`` removes the last digit.
- ✓ `` numb * 10 + remainder`` builds the reversed number.
- ✓ Compare the original number with its reverse.
- ✓ ``12321`` is a Palindrome Number.

Thank you 😊